

Unit V: Fourier Transforms

Fourier Transform

Many engineering problems lead to ordinary or partial differential equations which have to be solved under various types of conditions formulated from the problem. We are already familiar with the solution of higher order ordinary differential equations with initial conditions (initial value problems) using Laplace transforms. Solution of some partial differential equations with boundary conditions (boundary value problems) can be obtained with the help of Fourier transforms. Fourier transform is a mathematical tool for representing signals in time domain and frequency domain. It converts a time domain signal to frequency domain. It is an extension of Fourier series to non-periodic signals. Fourier transform is used in communications engineering as well as image and signal processing.

Definition: The Fourier Transform of $f(t)$ or complex Fourier transform of $f(t)$ is defined as $F(\omega)$

$$\mathcal{F}[f(t)] = F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$

The inverse Fourier transform of $f(t)$ is given by,

$$\mathcal{F}^{-1}\{\mathcal{F}[f(t)]\} = \mathcal{F}^{-1}\{F(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

Special Case of Laplace transform:

By the definition of bilateral Laplace transform

$$L[f(t)] = F(s) = \int_{-\infty}^{\infty} e^{-st} f(t) dt$$

where $s = \sigma + i\omega$.

If we set $\sigma = 0$, we obtain

$$F(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt$$

which is the Fourier transform of $f(t)$.

Unit V: Fourier Transforms

Properties of Fourier Transform:

1. Linearity property: If a and b are any 2 constants and $f(t)$, $g(t)$ are any 2 functions whose Fourier transforms exist, then

$$\mathcal{F}[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)]$$

Proof: By definition of Fourier transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\mathcal{F}[a f(t) + b g(t)] = \int_{-\infty}^{\infty} \{a f(t) + b g(t)\} e^{-i\omega t} dt$$

$$\mathcal{F}[a f(t) + b g(t)] = a \left[\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right] + b \left[\int_{-\infty}^{\infty} g(t) e^{-i\omega t} dt \right]$$

$$\therefore \mathcal{F}[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)] = a F(\omega) + b G(\omega)$$

2. Shifting property: If a is a real constant then

$$\text{a) Time shifting property: } \mathcal{F}[f(t - a)] = e^{-i\omega a} F(\omega)$$

Proof:

$$\mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\mathcal{F}[f(t - a)] = \int_{-\infty}^{\infty} f(t - a) e^{-i\omega t} dt$$

$$\text{Let } s = t - a \Rightarrow t = s + a$$

$$ds = dt$$

If $t \rightarrow -\infty, s \rightarrow -\infty$ and if $t \rightarrow \infty, s \rightarrow \infty$

Unit V: Fourier Transforms

$$\therefore \mathcal{F}[f(t-a)] = \int_{-\infty}^{\infty} f(s) e^{-i\omega(a+s)} ds$$

$$\mathcal{F}[f(t-a)] = \int_{-\infty}^{\infty} f(s) e^{-i\omega a - i\omega s} ds$$

$$\mathcal{F}[f(t-a)] = \int_{-\infty}^{\infty} f(s) e^{-i\omega a} e^{-i\omega s} ds$$

$$\therefore \mathcal{F}[f(t-a)] = e^{-i\omega a} \left[\int_{-\infty}^{\infty} f(s) e^{-i\omega s} ds \right] = e^{-i\omega a} F(\omega)$$

b) Frequency shifting property: $\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$

Proof:

$$\mathcal{F}[f(t)e^{iat}] = \int_{-\infty}^{\infty} f(t) \cdot e^{iat} \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)e^{iat}] = \int_{-\infty}^{\infty} f(t) \cdot e^{iat - i\omega t} dt = \int_{-\infty}^{\infty} f(t) \cdot e^{-i(\omega - a)t} dt$$

$$\therefore \mathcal{F}[f(t)e^{iat}] = F(\omega - a)$$

3. Modulation Theorem

If $\mathcal{F}[f(t)] = F(\omega)$ and ω_0 is any real number, then

$$\mathcal{F}[f(t) \cos(\omega_0 t)] = \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)]$$

and

$$\mathcal{F}[f(t) \sin(\omega_0 t)] = \frac{i}{2} [F(\omega + \omega_0) - F(\omega - \omega_0)]$$

Proof w.k.t $f(t) \cos(\omega_0 t) = \frac{f(t)}{2} [e^{i\omega_0 t} + e^{-i\omega_0 t}]$

$$\mathcal{F}\left[\frac{f(t)}{2} [e^{i\omega_0 t} + e^{-i\omega_0 t}]\right] = \frac{\mathcal{F}[f(t)[e^{i\omega_0 t}]]}{2} + \frac{\mathcal{F}[f(t)[e^{-i\omega_0 t}]]}{2} = \frac{F(\omega - \omega_0)}{2} + \frac{F(\omega + \omega_0)}{2}$$

If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$ (frequency shift property)

Unit V: Fourier Transforms

4. Duality property or symmetry property: If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

Proof: By the definition of inverse Fourier transform,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

Replacing $t \rightarrow -t$

$$2\pi f(-t) = \int_{-\infty}^{\infty} F(\omega) e^{-i\omega t} d\omega$$

Interchanging t and ω

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt = \mathcal{F}[F(t)]$$

5. Fourier transform of derivative:

Let $f(x)$ be continuous on the x -axis and $f(t) \rightarrow 0$ as $t \rightarrow \infty$, then

$$\mathcal{F}[f'(t)] = i\omega \mathcal{F}[f(t)]$$

Proof:

$$\mathcal{F}[f'(t)] = \int_{-\infty}^{\infty} f'(t) \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f'(t)] = \left\{ e^{-i\omega t} f(t) \right\}_{-\infty}^{\infty} - (-i\omega) \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt$$

$$\mathcal{F}[f'(t)] = 0 + i\omega \mathcal{F}[f(t)] = i\omega \mathcal{F}[f(t)]$$

Similarly, $\mathcal{F}[f^n(t)] = (i\omega)^n \mathcal{F}[f(t)]$

6. Differential with respect to frequency (ω)

If $\mathcal{F}(f(t)) = F(\omega)$ then

$$\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$$

Proof: Differentiating with respect to ω

$$F'(\omega) = \frac{d}{d\omega} \left(\int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt \right)$$

$$F'(\omega) = \int_{-\infty}^{\infty} \frac{d}{d\omega} (f(t) \cdot e^{-i\omega t}) dt$$

Unit V: Fourier Transforms

$$F'(\omega) = \int_{-\infty}^{\infty} -it \cdot f(t) \cdot e^{-i\omega t} dt$$

$$F'(\omega) = -i[\mathcal{F}(tf(t))]$$

Similarly

$$F''(\omega) = (-i)^2[\mathcal{F}(t^2f(t))]$$

Therefore $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$

$$(-i)^n t^n f(t) = \mathcal{F}^{-1}(F^{(n)}(\omega))$$

Fourier transform of basic functions:

1. Unit impulse function, $f(t) = \delta(t)$

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-i\omega t} dt = [e^{-i\omega t}]_{t=0} = 1$$

$\therefore$ Impulse in time $\Rightarrow$ constant in frequency

2. Constant function, $f(t) = k$

By Duality property, we have

$$\text{If } \mathcal{F}[f(t)] = F(\omega) \text{ then } \mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$\text{We know that } \mathcal{F}[\delta(t)] = 1$$

$$\Rightarrow \mathcal{F}[1] = 2\pi\delta(-\omega) = 2\pi\delta(\omega) \quad (\because \text{impulse function is even})$$

$$\therefore \mathcal{F}[k] = 2\pi k\delta(\omega)$$

Hence constant in time $\Rightarrow$ Impulse in frequency

3. $f(t) = e^{iat}$

By frequency shifting property, we have

$$\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$$

Unit V: Fourier Transforms

$$\text{Also } \mathcal{F}[1] = 2\pi\delta(\omega)$$

$$\therefore \mathcal{F}[1 \cdot e^{iat}] = \mathcal{F}[e^{iat}] = 2\pi\delta(\omega - a)$$

4. Trigonometric functions, $f(t) = \cos at$ and $f(t) = \sin at$

$$\mathcal{F}[\cos at] = \mathcal{F}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2}\{\mathcal{F}[e^{iat}] + \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\cos at] = \frac{1}{2}\{2\pi\delta(\omega - a) + 2\pi\delta(\omega + a)\}$$

$$\therefore \mathcal{F}[\cos at] = \pi\delta(\omega - a) + \pi\delta(\omega + a) = \pi\{\delta(\omega - a) + \delta(\omega + a)\}$$

$$\mathcal{F}[\sin at] = \mathcal{F}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i}\{\mathcal{F}[e^{iat}] - \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\sin at] = \frac{1}{2i}\{2\pi\delta(\omega - a) - 2\pi\delta(\omega + a)\} = \frac{1}{i}\{\pi\delta(\omega - a) - \pi\delta(\omega + a)\}$$

$$\therefore \mathcal{F}[\sin at] = -i\pi\delta(\omega - a) + i\pi\delta(\omega + a) = i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$$

Problems

1. Find Fourier transform of $f(t) = e^{at}u(-t)$.

Solution :

Consider $g(t) = e^{-at}u(t)$ and $\mathcal{F}[g(-t)] = G(-\omega)$

$$\mathcal{F}[g(t)] = \frac{1}{a+i\omega} = G(\omega)$$

$$G(-\omega) = \frac{1}{a-i\omega} = \mathcal{F}[e^{at}u(-t)]. \text{ since } f(t) = g(-t)$$

$$F(\omega) = G(-\omega)$$

2. Find Fourier transform of $y(t) = e^{-a|t|}$.

Solution

Unit V: Fourier Transforms

given input can be written in terms of unit step function

$$y(t) = e^{-at}u(t) + e^{at}u(-t)$$

$\mathcal{F}[y(t)] = \mathcal{F}[e^{-at}u(t) + e^{at}u(-t)]$ by Linearity property

$$= \mathcal{F}[e^{-at}u(t)] + \mathcal{F}[e^{at}u(-t)] \text{ time reversal property}$$

$$= \frac{1}{a+i\omega} + \frac{1}{a-i\omega} = \frac{2a}{a^2+\omega^2}$$

3. Find the inverse Fourier transform of $\frac{e^{3i\omega}}{2-i\omega}$

$$\text{wkt } \mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$$

$$e^{-at}u(t) = \mathcal{F}^{-1}\left[\frac{1}{a+i\omega}\right] \quad \mathcal{F}[e^{at}u(-t)] = \frac{1}{a-i\omega}.$$

$$\mathcal{F}^{-1}\left[\frac{1}{2-i\omega}\right] = e^{2t}u(-t) = f(t)$$

$$\mathcal{F}^{-1}[e^{-i\omega a}F(\omega)] = f(t-a)$$

$$\mathcal{F}^{-1}\left[\frac{e^{3i\omega}}{2-i\omega}\right] = f(t+3) = e^{2(t+3)}u(-(t+3))$$

5. Solve differential equation using Fourier transform

$$f' - 4f = u(t)e^{-4t}$$

$$i\omega F(\omega) - 4F(\omega) = \frac{1}{4+i\omega}$$

$$(i\omega - 4)F(\omega) = \frac{1}{4+i\omega}$$

$$F(\omega) = \frac{1}{(4+i\omega)(i\omega - 4)}$$

$$F(\omega) = \frac{-1}{(4+i\omega)(4-i\omega)}$$

$$F(\omega) = -\frac{1}{4^2 + \omega^2} \quad \left\{ \frac{2a}{a^2 + \omega^2} = \mathcal{F}[e^{-a|t|}] \right\}$$

$$f(t) = -\frac{1}{2 \cdot 4} e^{-4|t|} = -\frac{1}{8} e^{-4|t|}$$

Unit V: Fourier Transforms

6. Find Fourier transform of $\frac{1}{2\pi} \text{sinc} t$

Solution

We know that $F(\omega) = \frac{1}{\sqrt{2\pi}} \text{sinc} \omega$ for

$$f(t) = \begin{cases} 0 & t < -1 \\ 1 & -1 < t < 1 \\ 0 & t > 1 \end{cases} \quad \text{Typical box wave}$$

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$\mathcal{F}\left(\frac{1}{\sqrt{2\pi}} \text{sinc} t\right) = \mathcal{F}(F(t)) = 2\pi f(-\omega) = \begin{cases} 0 & -\omega < -1 \\ 2\pi & -1 < -\omega < 1 \\ 0 & -\omega > 1 \end{cases}$$

7. Obtain Fourier transform for $[te^{-4t}u(t)]$.

Solution: w.K.t $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$

And from previous property $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$

$$\mathcal{F}[e^{-4t}u(t)] = \frac{1}{4+i\omega}$$

$$\mathcal{F}[te^{-4t}u(t)] = i \cdot \frac{d}{d\omega} \left[\frac{1}{4+i\omega} \right] = i \left[-\frac{i}{(4+i\omega)^2} \right] = \frac{1}{(4+i\omega)^2}$$

Unit V: Fourier Transforms

The Fourier cosine transform of a real function is the real part of the full complex Fourier transform

RECALL

If $f(t)$ be defined in $0 \leq t < \infty$ and $f(t)$ is extended as even function in $(-\infty, \infty)$ satisfying the condition on Fourier integral theorem. The Fourier cosine transform for the function $f(t)$ is defined by

$$F_c(\omega) = \int_0^{\infty} f(t) \cos \omega t \, dt$$

Where ' ω ' is a parameter

The Fourier sine and cosine transforms are forms of the Fourier transform that do not use complex numbers or require negative frequency. They are the forms originally used by Joseph Fourier and are still preferred in some applications, such as signal processing or statistics

DEFINITION

INVERSE FOURIER TRANSFORM

If $F_c(\omega)$ is the Fourier cosine transform of $f(t)$ which satisfies Dirichlet's condition in every finite interval $(0, l)$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos \omega t \, d\omega$$

$f(t)$ is called inverse Fourier cosine transform of $F_c(\omega)$

Unit V: Fourier Transforms

1. Find the inverse Fourier cosine transform of $e^{-\omega}$

$$\mathcal{F}_c(\omega) = e^{-\omega}$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos \omega t d\omega$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} e^{-\omega} \cos \omega t d\omega$$

$$\text{w.k.t } \int_0^{\infty} e^{-a\omega} \cos \omega t d\omega = \frac{a}{a^2 + t^2} \text{ and } \int_0^{\infty} e^{-a\omega} \sin \omega t d\omega = \frac{t}{a^2 + t^2}$$

$$f(t) = \frac{2}{\pi} \left(\frac{1}{1^2 + t^2} \right)$$

2. Find $f(t)$ if its Fourier cosine transform is $\mathcal{F}_c(\omega) = \begin{cases} k - \frac{\omega}{2} & \text{for } \omega < 2k \\ 0 & \text{for } \omega > 2k \end{cases}$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos \omega t d\omega$$

$$f(t) = \frac{2}{\pi} \int_0^{2k} \left(k - \frac{\omega}{2} \right) \cos \omega t d\omega + \int_{2k}^{\infty} (0) \cos \omega t d\omega$$

Integration by parts

$$f(t) = \frac{2}{\pi} \left[\left(k - \frac{\omega}{2} \right) \frac{\sin \omega t}{t} \right]_{\omega=0}^{\omega=2k} - \left[\left(-\frac{1}{2} \right) \frac{-\cos \omega t}{t^2} \right]_{\omega=0}^{\omega=2k}$$

$$f(t) = \frac{2}{\pi} \left[\left[\left(k - \frac{2k}{2} \right) \frac{\sin 2kt}{t} - \left(k - \frac{0}{2} \right) \frac{\sin 0}{t} \right] - \frac{1}{2} \left[\frac{\cos 2kt}{t^2} - \frac{\cos 0}{t^2} \right] \right]$$

$$f(t) = \frac{2}{\pi} \left[\frac{-1}{2} \left[\frac{\cos 2kt}{t^2} - \frac{1}{t^2} \right] \right]$$

Unit V: Fourier Transforms

$$f(t) = \frac{1}{\pi} \left[\frac{2\sin^2 kt}{t^2} \right]$$

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